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1 **Ice storms as a successional pathway for *Fagus grandifolia* advancement in *Quercus rubra***
2 **dominated forests of Southern New England**

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Abstract

On December 11th, 2008 a severe ice storm affected large portions of Southern New England. We report the results of a study investigating differential damage among tree species. Past studies surveying ice damaged forests have relied heavily on ocular estimations of canopy damage. We compare this method to damage estimates based on the cross-sectional area of downed woody material and quantitative comparisons of tree height and canopy projection in damaged and undamaged stands. Ocular estimates and changes in canopy height and projection were unreliable. Estimates based on the amount of woody debris provide a more robust measure of storm damage. We assess damage in two forest tracts with a dominant red oak (*Quercus rubra* L.) canopy (42% of total basal area), and an American beech (*Fagus grandifolia* Ehrh.) dominated understory (50% of all stems >1.3 m). Species' standing basal area was well correlated with the amount of newly downed woody debris ($r=.69$, $p<.001$); accordingly, oak species constituted approximately 57% of all newly downed woody debris, and ANOVA showed oak species were significantly more impacted by ice damage than other species present ($p<.001$). Our findings indicate that canopy species provide an "umbrella effect", shielding less dominant lower strata trees from the worst effects of ice load. Oak was also the largest contributor to older downed woody debris, indicative of a recurrent historical pattern of differential canopy disturbance driven by strata and relative dominance. These findings suggest that periodic ice events play an important role in facilitating the advancement of shade tolerant tree species into the canopy from lower strata.

Keywords: glaze, forest succession, forest dynamics, disturbance, nectria, red oak, beech.

Introduction

Ice storms regularly cause widespread canopy damage to northern forests. While there is wide agreement that ice damage significantly affects forests, the implications for stand development and successional process are not well understood (Whitney and Johnson 1984; Lafon 2004; 2006). During an ice storm, ice accumulates as rain falls through a layer of supercooled air and freezes upon impact with subfreezing surfaces such as stems and branches (Lemon 1961). Ice storms are far more common than any other temperate forest canopy disturbances with reported return intervals ranging from just 5 years (Bragg et al. 2003), to 20-100 years (Smith 2000). The Great Mountain Forest meteorological station in Northeastern Connecticut, recorded 15 separate ice events in 1952 alone (Childs 1953). During extreme events as much as 90 mm of ice may accumulate (Proulx and Greene 2001). This additional weight interacts with factors such as windspeed, slope and tree architecture (Bruederle and Stearns 1985; Rhoads et al. 2002), and results in broken branches, snapped crowns and fully uprooted trees (Carvell et al. 1957). Falling limbs and trees may also lead to severe damage not directly related to ice load (Boerner et al. 1988). Many studies have looked at variation in susceptibility to ice storm damage across species, but there is little consistency in their findings (Boerner et al. 1988; Seischab et al. 1993; Van Dyke 1999). The amount of accumulated ice and the size of the tree appear to be related to the amount of damage (Proulx and Greene 2001), though other seemingly relevant factors such as species growth morphology, stand position, and canopy position vary widely in their effects on tree damage (Van Dyke 1999).

Predictions regarding the effects of ice storms on patterns of forest succession also vary widely. In some cases authors predict that ice damage slows succession, favoring intolerant pioneer species and reducing tree species diversity (Boerner et al. 1988). Other studies suggest an increase in stand-level species diversity as a result of accelerated successional process (Lemon 1961; Proulx and Greene 2001; Rhoads et al. 2002). Predictions about future forest development following ice damage should be based on stand-specific understanding of pre-disturbance composition, structure, developmental history and trajectory, and an empirical assessment of damage severity.

Ice storms dramatically affect American beech (*Fagus grandifolia* Ehrh.) (Melancon and Lechowicz 1987), though the implications of that damage for future stand development are less clear. One study investigating the impacts of an extreme ice event in northern hardwoods predicted that beech would respond vigorously following an intense storm (Rhoads et al. 2002). The amount of crown damage and the resulting sprouting may both be more extreme given the prominent role of beech bark disease—the fungal-scale complex caused by the interaction of *Nectria coccinea* var. *faginata* and *Cryptococcus fagisuga*. Beech infected by beech bark disease are more prone to breakage in storms (Houston 1983) but maintain their ability to sprout vigorously in response to disturbance, suggesting diseased beech will remain a significant component of the forest after disturbances (Takahashi et al. 2010). Ice storm damage to oak (*Quercus*) species has been quantified in general terms for certain eastern forest types (Rebertus et al. 1997; Van Dyke 1999), but there has been little discussion of oak-beech forests' responses to ice storms (Hooper et al. 2001).

To date there are no studies investigating the implications of differential damage among species following moderate ice storm events on the development of even-aged second-growth forests that feature an oak canopy and beech understory. The oak-beech forest type is widespread in eastern North America, particularly within the Appalachian Mountains (Braun 1950; Eyre 1980); and it is also found at middle elevations in association with mountains across Central Europe (Cajander 1949), and Northeast Asia (Kolbek et al. 2003). This study adds to our understanding of developmental process following a common forest disturbance by quantifying damage patterns in ice affected forest stands. Our main objective is to predict how ice storms drive succession in these forests. We hypothesize that ice events serve as an important successional pathway for shade tolerant beech to ascend into the canopy.

Methods and Materials

Site description

Measurements of stand composition, canopy damage, and downed woody debris were taken in the Berkshire Mountains, an extension of the Appalachian range, between July and October 2009 at the Great

Mountain Forest in Norfolk, CT (latitude 41 ° 58 ' N; longitude 73° 13' W) and at the Liberman Forest in Sandisfield, MA (latitude 42 ° 03 ' N; longitude 73° 06' W (Fig. 1). The two forests are 17 kilometers apart and are similar in terms of climate and topography. Elevations at study sites range between 425 m and 525 m. As with much of Southwestern New England, these forests were cleared for charcoal production between 1800-1890 and largely abandoned around 1890-1930. This legacy of intense anthropogenic disturbance shaped the oak-beech coppice stands that continue to characterize the landscape (Egler 1940; Winer 1955).

The December 11, 2008 ice storm caused extensive damage through a large section of Southern New England (Abel and Ellement 2008). Damage from the storm at both forests was largely constrained to east and southeast facing slopes and concentrated at an elevation of approximately 460 m. Surveyed sites were located at the southern extent of the storm, and damage in the study areas was generally less severe than in areas 40 km to the north and 65 km to the east.

Winer (1955) provides a detailed discussion of the Great Mountain Forest's soils, climate, and forest type that is largely applicable to the Liberman Forest. The Great Mountain forest plots are on Westminster-Millsite-Rock outcrop complex soils while, the Liberman Forest plots are on stony Lyman-Tunbridge and Berkshire-Marlow soils (NRDC 2007). Surveyed stands vary in age from 70-120 years, and contain a mixture of hardwoods with a dominant red oak (*Quercus rubra* L.) canopy, numerous under- and mid-story beech stems, and minor elements of eastern hemlock (*Tsuga canadensis* (L.) Carr.) and white pine (*Pinus strobus* L.) (Fig. 2). Beech bark disease in a moderate-to-severe infestation was present in all stands, along with an abundance of sprout origin beech saplings.

Sampling Design and Data Collection

Based on topography, elevation, aspect and a preliminary survey we identified a total of eight separate ice-affected stands; four stands were delineated at Great Mountain Forest and four at the Liberman Forest. These impacted stands were paired with 5 control stands—4 in the Liberman forest and 1 in the Great Mountain

Forest. Only one control stand was assessed at Great Mountain because ice storm damage was widespread, leaving no unaffected areas of the same species composition, and age.

Within each stand we randomly located 3 points to serve as the centers of 12 m radius fixed-area plots. For each plot—a total of 24 damaged, and 14 control—we recorded species, diameter at breast height, total height, bole height, and canopy projection for all stems ≥ 5 cm at DBH (1.3 m). For these large stems we followed Rhoads et al. (2002) and assigned a level of ice storm damage with 0=no damage, 1= 0-20% damage, 2= 20-40% damage, 3=40-60% damage, 4= 60-80% damage and 5= 80-100% of crown or bole damage. We recorded species, total canopy height, bole length, and damage class for all stems < 5 cm. Heights were measured to an accuracy of ± 2 m with the use of an Opti-Logic ‘Insight’ laser hypsometer. Following Gregoire and Valentine (1995), we estimated crown projection by measuring from each stem to the edge of the live crown on a random azimuth. Canopy height, bole length, and crown projection were used to estimate live crown ratio.

In addition for each of the 38 sample plots, two 30 m line transects were established at right angles originating from each plot center. We measured all pieces of downed wood with a base diameter ≥ 3 cm intersecting each transect and recorded species, and temporal provenance (newly downed vs. historical). For the purpose of estimating abundance, the perpendicular length of each piece was also recorded.

Data Analysis

Patterns in ice storm damage are driven by both topography and aspect (Rhoads et al. 2002), making locating unaffected controls in similar topographic position, aspect, and correspondingly species composition difficult. In addition, the efficacy of using ocular estimates of percent damage to individual trees is questionable (Innes 1993; Van Dyke 1999). To account for the inherent variability in damage with changes in site and for the subjective nature of damage estimations we used two techniques (Fig. 3). In the first we quantified the damage to individuals by recording the presence of broken and hanging branches and by estimating reductions in live crown ratio and canopy projection. In addition to the individual tree assessments, we recorded whether downed

woody debris was fresh (i.e. from this ice storm), or from prior events. Differences between debris that originated from the December 2008 ice storm (material that was less than a year old), and material that had fallen in prior years from storms was easily distinguishable. We then estimated the total amount of downed woody material to assess relative damage between species. Cross-sectional branch area is a strong indicator of attached leaf area and woody biomass (Hooper et al. 2001; Waring et al. 1982), and consequently the amount of live crown removed by storm damage. This technique, while not entirely novel, has been used relatively few times to assess damage among species following ice events (Bruederle and Stearns 1985; Hooper et al. 2001). In addition to being a more robust method of damage estimation, the comparison of legacy and recent woody debris has the added benefit of providing baseline, *post hoc*, control and eliminating the known conflation of site factors in control comparisons.

Following Gregoire and Valentine (1995) the canopy projection of each tree was estimated using a single radius measured from the bole of each tree at a random azimuth. Cross-sectional area of woody debris was estimated for each species using data from the line intersect sampling transects, and formulae adapted from (Gregoire and Valentine 2008):

Eq. (1)
$$Total\ Cross\ Sectional\ Area = \frac{\sum y_k}{\pi_k(\theta_x)}$$

Where y_k is the cross sectional area of an individual piece k and π_k is the inclusion probability given by:

Eq. (2)
$$\pi_k(\theta_x) = \frac{w_k L}{A}$$

Where w_k is the length of piece k measured perpendicularly to transect, A is the area of the stand, and L is transect length.

Relationships between measured variables were assessed using Pearson's test for correlation. We used ANOVA with species and control groups as factors to examine species contribution to downed woody debris pools (CSAWD, newly downed and historical). Similar comparisons were made between CSAWD across control groups (damaged vs. undamaged stands). Because a total of 25 comparisons were made, we applied a Bonferonni correction for multiple comparisons (Dunn 1961) and report only p-values of less than .002 as significant. All analyses were conducted using the statistical freeware R (R Development Core Team, 2010).

Results

Attempts to quantify differential damage among species by measuring reductions in live crown ratio and canopy projections showed no significant differences between impacted and control stands. Ocular estimates of canopy damage were not correlated with the cross-sectional area of downed woody debris (CSAWD) ($r=0.19$, $p=0.12$) (Fig 4). The number of stems per hectare was not well correlated with newly downed debris ($r=-0.02$, $p=0.86$). Tree height was correlated with newly downed debris, though the relationship was only marginally significant ($r=.23$, $p=.07$). In contrast, the estimated cross-sectional area of newly downed woody debris showed a significant correlation to standing basal area ($r=.69$, $p<.001$) and there was a significantly greater CSAWD contributed by canopy dominant red oak ($p<.001$) as compared to other species.

Oak species were by far the most impacted, comprising nearly 57% of the total cross-sectional area of newly downed woody debris. In contrast, American beech accounted for only a small proportion of the total ice related CSAWD (7%). Additionally, upper strata species were more likely to be found in large quantities in the recent debris pool. Taken together oak, white ash (*Fraxinus americana* L.), black cherry (*Prunus serotina* Ehrh.), and white pine represented 80% of all newly downed CSAWD. Eastern hemlock was not greatly impacted by the storm and represented a very small portion of newly downed woody debris (~0.25%). There was a general trend of increasing CSAWD with increasing relative basal area among species, although white pine was disproportionately damaged whereas red maple was disproportionately spared (Fig 4).

Oak species comprised a large portion of the historical cross-sectional area of downed woody debris as well ($3.70 \pm 0.79 \text{ m}^2\text{ha}^{-1}$, mean $\pm$ SE); however, the proportion of oak in the historic CSAWD was less than in newly fallen material (20% vs. 57%). Beech was a larger component of the historical CSAWD but not significantly so ($p>0.9$). While white ash was only a minor component of ice debris, it was a far more prominent component of the historical CSAWD comprising just over 20%. Black Cherry was also a relatively large component (14%) of the historical CSAWD. Taken together, paper birch (*Betula papyrifera* Marsh), white oak (*Quercus alba* L.), and relic American chestnut (*Castanea dentata* Marsh) accounted for just 8% of historical CSAWD (Table 1). Though still significant, the relationship between historic debris pool and basal area ($r=0.39$, $p<.001$) was less compelling than the same relationship in the ice-related pool. Stem density was not correlated with the historic CSAWD ($r=0.12$, $p=0.34$).

A comparison of ocular estimates showed trees in control stands were significantly less damaged ($p<0.001$) than those in ice-affected stands. The $0.35 \pm .02 \text{ m}^2\text{ha}^{-1}$ of newly fallen CSAWD in undamaged stands was less than new CSAWD found in damaged stands (mean difference= $1.33 \text{ m}^2\text{ha}^{-1}$ $p<.005$). The largest contributor to new CSAWD in control stands was white pine, though the amount was still less than found in damaged stands. There was also less historical CSAWD present in undamaged stands (mean difference= $1.24 \text{ m}^2\text{ha}^{-1}$ $p<.001$). Understory species contributed 33% of the historical CSAWD in undamaged stands with beech, red oak, and white pine comprising approximately another third. In contrast to ice damaged stands, standing basal area in control stands was poorly correlated with both new ($r=-0.05$, $p=0.7311$) and historical ($r=0.09$, $p=0.58$) CSAWD; the number of stems was also uncorrelated ($r=-0.10$, $p=0.50$) with the CSAWD.

Discussion

There were clear patterns in the damage among species in these mixed-species stratified forests in southern New England. In general, canopy species were more heavily damaged. Oak species—and where present, black cherry and white pine—was more severely impacted than later successional, and lower strata

counterparts, particularly beech and hemlock. These patterns relate to relative basal area and current canopy position.

Estimated canopy projections and measured tree heights between trees in damaged and control stands did not differ. Even when damage was readily apparent, remaining branches made objectively quantifying canopy reduction based on comparisons of height and canopy projection extremely challenging (Fig. 3). While this may be possible with much larger datasets in stands with more severe damage, for finer-scale measurements, its practicality is questionable. Similarly, the lack of stand level correlation between subjective ocular estimates of damage and estimates of the CSAWD suggest that ocular estimation does not produce a reliable measure of storm damage.

Our results are consistent with others who have found upper strata canopy position (Boerner et al. 1988; Bragg et al. 2003; Bruederle and Stearns 1985; Carvell et al. 1957; Hooper et al. 2001; Rhoads et al. 2002) and early successional status (Lafon 2006; Lemon 1961; Seischab et al. 1993; Van Dyke 1999) to be important factors driving susceptibility to ice damage. This is in contrast to studies suggesting that tolerance or susceptibility could be categorized by species. The general lack of consistency in species-specific ratings of susceptibility calls into question their usefulness. For example, Van Dyke (1999) notes various studies rating beech, oak, hemlock and sugar maple (*Acer sacharumm*. Marsh) as being of either low, intermediate, or high susceptibility to damage from ice load. Given these inconsistencies, and the data presented here, it appears likely that stratification pattern at given successional stage plays a larger role in determining differential damage than does species-specific tolerances to ice loading.

The exceptions to the general trend correlating relative basal area and proportion of the ice debris pool were the relatively high contribution of white pine and the relatively low contribution of red maple. While not statistically significant here, the observed discrepancy is consistent with ice functioning as a canopy level disturbance with white pine in the emergent strata and red maple in the middle strata (Liptzin and Ashton 1999; Tift and Fajvan 1999). White pine was only a minor component of undamaged stands (1%); however it contributed (85%) of new debris found there. This implies that even in stands not significantly impacted by

glaze, the emergent and evergreen white pine is particularly susceptible to ice damage. By contrast, in damaged stands, red maple accounted for just 5% of newly downed woody debris, despite being a relatively large component of total basal area (12%). This is likely explained by relative canopy position. In these stands red maple largely occupies a subcanopy position—as is common in northeastern forests (Lorimer 1984)—potentially shielding it from the worst ice loading.

Takahashi et al. (2010) concluded that ice storms serve to maintain the relative abundance of beech in mixed-species forests, while others have concluded that beech is more susceptible to ice damage when infected with beech bark disease (Latty 2005). Our results indicate the potential for a dramatic increase in the relative abundance of beech as the result of the recurrence of episodic ice storms in oak-beech forests; it suffers little damage, and when impacted, its propensity for sprouting all but ensures its future place in the stand. The relatively low intensity of damage produced by moderate ice events also increases the likelihood that lower strata beech will preferentially benefit from glaze events (Melancon and Lechowicz 1987). Studies of understory light conditions following ice storm damage indicate that the gaps left after moderate ice damage can be insufficient to facilitate the development of a shade intolerant cohort (Beaudet et al. 2007). While these gaps may be too small to regenerate less tolerant species, gaps left by moderate storm damage are large enough to produce a substantial benefit to sub-canopy beech (Weeks et al. 2009). These favorable understory light conditions are complemented by root injury in the organic horizon, encouraging sprouting in beech (Jones and Raynal 1986). Copious sprouting is further facilitated by beech bark disease (Hane 2003; Houston 1994; Wagner et al. 2010). Further, few species are capable of regenerating under the dense thickets of beech (Hane 2003; Gravel et al. 2008; Latty 2005). Indeed, the regeneration ecology of beech is well tailored to relatively low intensity, frequent, canopy-level disturbance.

Several authors have addressed the potential successional implications of ice storms. Depending on the intensity of the disturbance, successional stage and species composition of the affected forest, ice storms may either accelerate or reset succession process (Boerner et al. 1988; Siccama et al. 1976; Whitney and Johnson 1984). The results of this study provide an example of conditions that are likely to produce the acceleration of

successional process: a canopy dominated by relatively shade intolerant species with an established understory of shade tolerant advance regeneration and smaller subcanopy trees awaiting release. In this forest system the predominance of beech in the understory is likely to lead to a decrease in species diversity and timber values. Increased frequency and intensity of ices storms associated with climate change may promote such shifts in species composition (Easterling, et al. 2000; Dale et al. 2001). Our results have wide scale application to oak-beech forests across eastern North America because of the common origin of oak-beech forests to this region from heavy cutting and agricultural abandonment in the late 19th and early 20th century (Whitney, 1996). The approximately 100 year old second-growth oak-beech forests that arose from this past land use history are at a successional stage and time where the oak dominated canopy is now susceptible to being overtopped by a shade tolerant beech sub-canopy positioned beneath it. Land use history of agricultural abandonment and forest cutting in northeast Asia (Totman 2014) and Central Europe (Arnould et al. 1997) associated with the industrial revolution that led to what is now similarly aged oak-beech forests make this study relevant to a much broader temperate forest type than the Appalachian mountain region eastern North America.

Conclusions

The use of woody debris in the estimation of storm damage is not entirely novel; however, it remains underutilized. This is especially true when researchers are concerned with understanding the relative damage patterns among a group of intimately mixed species. The applications of this technique are likely not limited to the interpretation of distinct exogenous disturbance; in the future, studies relating overstory structural patterns to relative abundance of coarse woody debris may provide valuable insight into the current and historical relationship between moderate intensity disturbances and developmental dynamics in mixed-species forests.

In the forest type examined in this study, canopy dominant oak species appear to provide an “umbrella effect” to trees in lower strata, effectively shielding them from the worst effects of glaze. The result is newly available growing space for relatively undamaged shade-tolerant species established in the lower strata. In this system, these strata are dominated by beech and while beech bark disease may weaken individual trees, they

will likely continue to advance to the canopy. We predict an increase in the relative abundance of beech as a result of this and similar ice events.

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Tables

Table 1. Woody Debris Pools

Damaged Stands

	New Debris (m ² ha ⁻¹) (+/-1SE)	% New Debris Pool	Historic Debris (m ² ha ⁻¹)(+/-1SE)	% Historic Debris Pool
Red Oak	8.63(2.81)	57%	3.70(0.79)	20%
White Pine	2.17(1.12)	14%	2.07(1.06)	11%
Beech	1.08(0.24)	7%	1.34(0.42)	7%
Red Maple	0.75(0.27)	5%	1.39(0.54)	7%
Ash	0.73(0.44)	5%	3.15(1.48)	17%
Cherry	0.65(0.25)	4%	0.23(0.22)	1%
Sugar Maple	0.56(0.17)	4%	0.14(0.07)	1%
Understory sp.	0.25(0.22)	2%	2.72(1.00)	1%
Birch	0.16(0.11)	1%	0.28(0.12)	14%
Hemlock	0.03(0.02)	1%	2.56(1.74)	1%
Other Overstory sp.	0.02(0.02)	0%	1.27(0.02)	8%
All Species	15.02(0.37)		18.85(0.28)	

Undamaged Stands

	New Debris (m ² ha ⁻¹) (+/-1SE)	% New Debris Pool	Historic Debris (m ² ha ⁻¹)(+/-1SE)	% Historic Debris Pool
Red Oak	0(0)	0%	0.78(0.36)	15%
White Pine	0.29(0.22)	85%	0.59(0.25)	11%
Beech	0(0)	0%	1.03(0.18)	19%
Red Maple	0.03(0.03)	8%	0.44(0.44)	9%
Ash	0(0)	0%	0(0)	0%
Cherry	0(0)	0%	0(0)	0%
Sugar Maple	0.02(0.02)	7%	0.03(0.03)	1%
Understory sp.	0(0)	0%	1.77(1.10)	33%
Birch	0(0)	0%	0.26(0.11)	5%
Hemlock	0(0)	0%	0.13(0.07)	3%
Other Overstory sp.	0(0)	0%	0.22(0.14)	4%
All Species	0.35(0.02)		5.22(0.13)	

Figure Legends

Figure 1. Study Area: The study sites are located in upland forests along the Massachusetts-Connecticut border in the towns of Sandisfield, Massachusetts and Norfolk, Connecticut. The Liberman Forest is located 17 km to the northeast of the Great Mountain Forest.

Figure 2. Plots of number of stems per hectare (upper) and basal area per hectare (lower) for both damaged (left) and undamaged (right) stands. Understory species include hornbeam (*Carpinus Carolinana* Walt.) , striped maple (*Acer pensylvanicum* L.) and hophornbeam (*Ostrya virginiana* Mill.); paper birch (*Betula papyrifera* Marsh.), American chestnut (*Castanea dentata* Marsh.) and white oak (*Quercus alba* L.) are grouped in the ‘Other Overstory’ category.

Figure 3. Assessing Canopy Damage. Large limbs remain on all but the most severely damaged trees meaning differences in total tree height and canopy projection are poor measures of storm damage across stands. Even when damage is severe, there may be little change in these metrics. Measuring woody debris pools better represent the impact of discrete storm events because it captures the loss of even relatively fine branches.

Figure 4. Percent of the cross-sectional area of downed woody material and relative basal area in damaged stands.

Figures

Figure 1

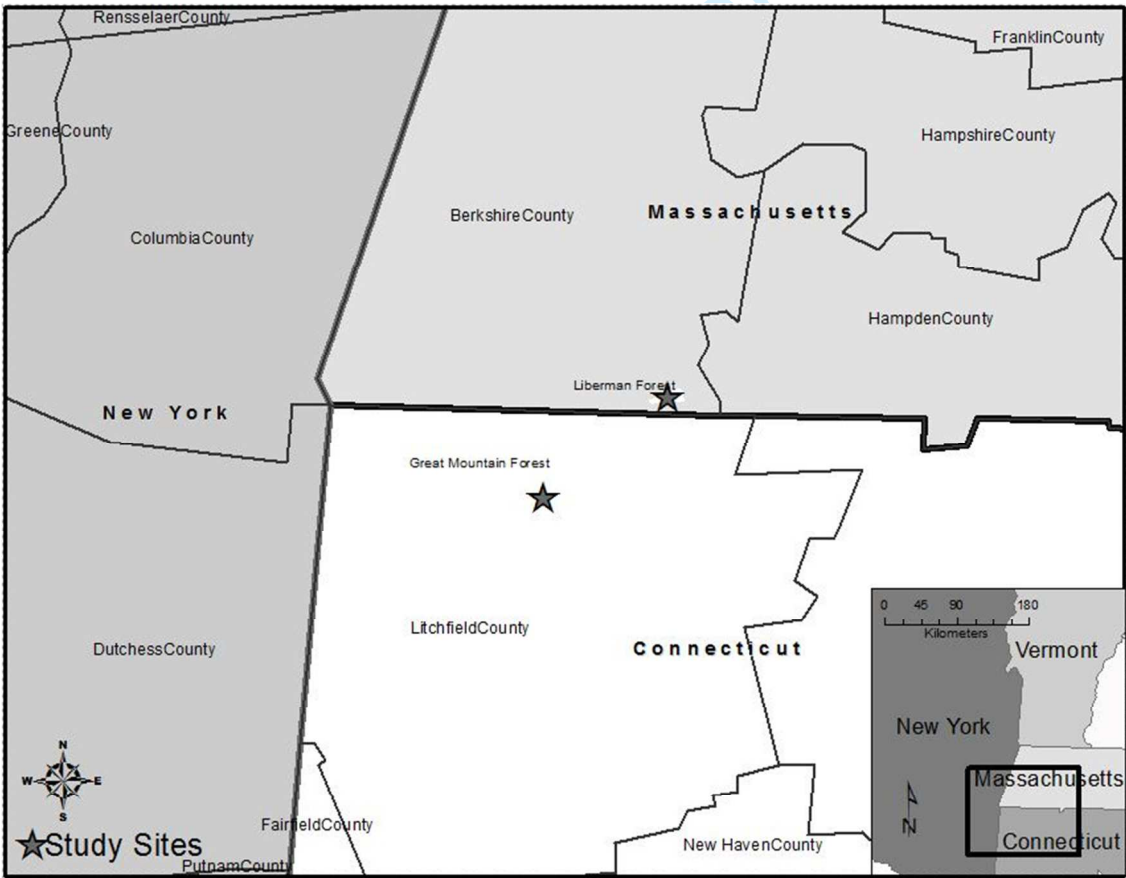


Figure 2

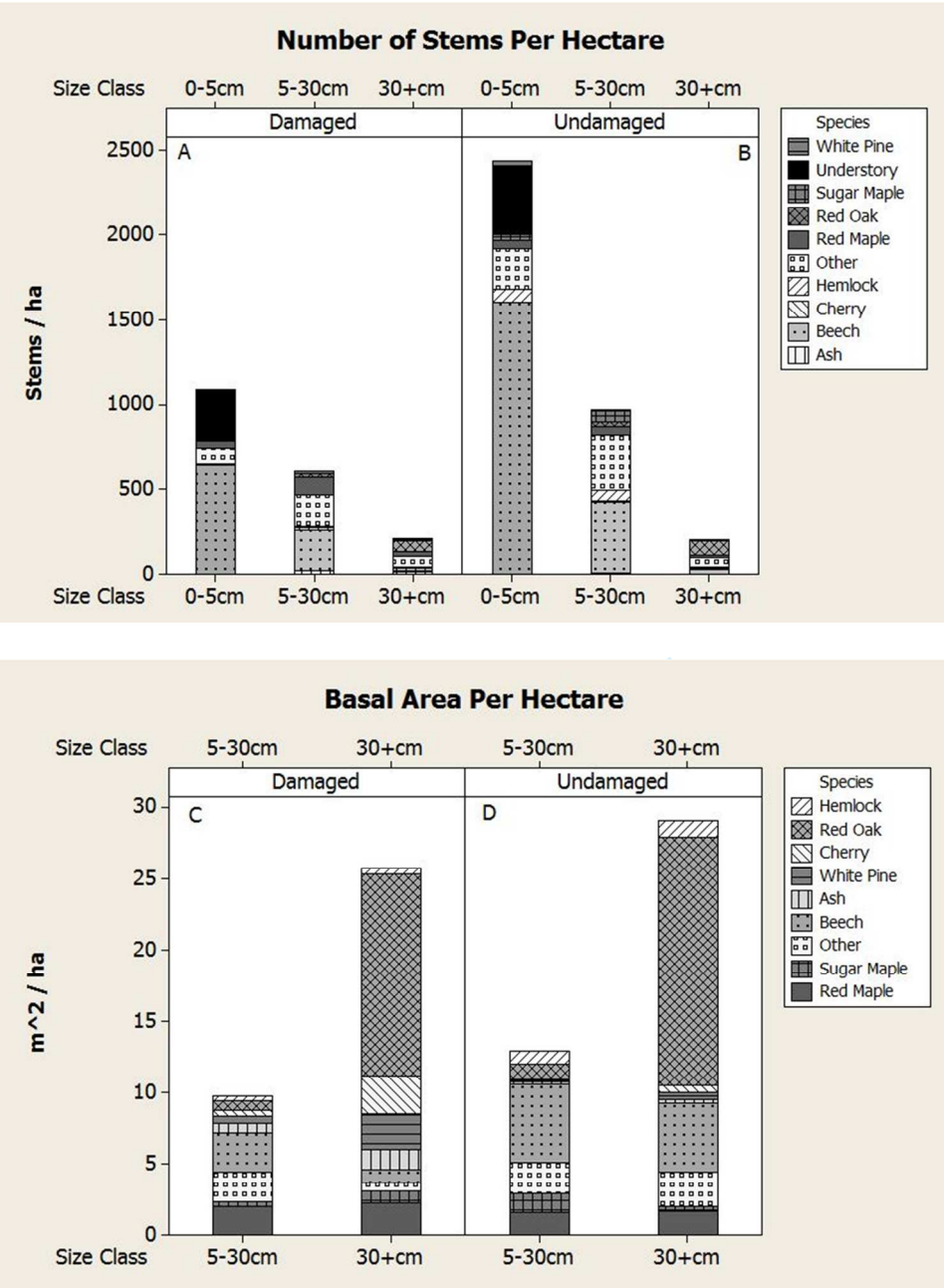


Figure 3

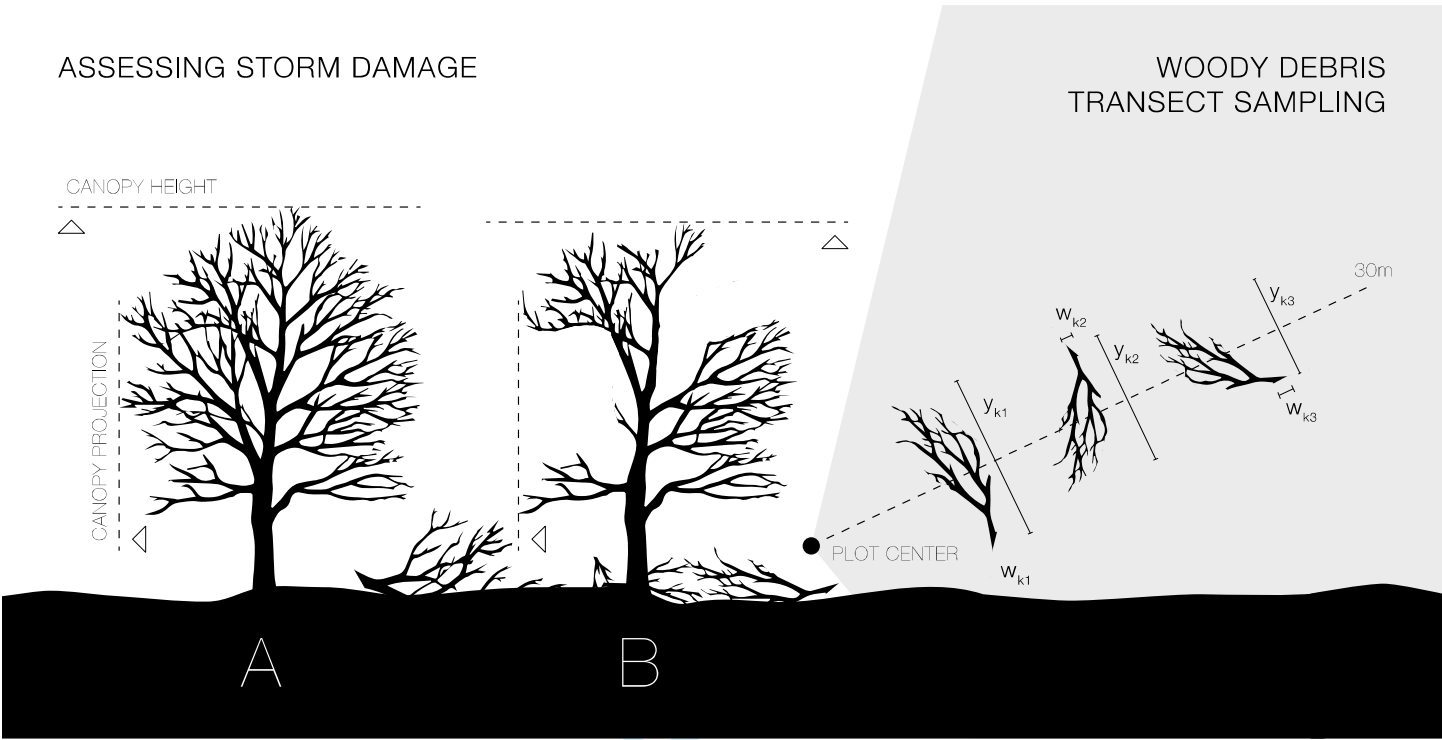


Figure 4

